RIGHT SPHERICAL TRIANGLE

SPHERICAL TRIGONOMETRY

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Cofunction Identities

Napier's Rule

Quadrantal Triangles

COFUNCTION IDENTITIES

COFUNCTION IDENTITIES

Basic Identities

$$\sin(\theta) = \cos(90° - \theta)$$

$$\cos(\theta) = \sin(90° - \theta)$$

$$\tan(\theta) = \cot(90° - \theta)$$

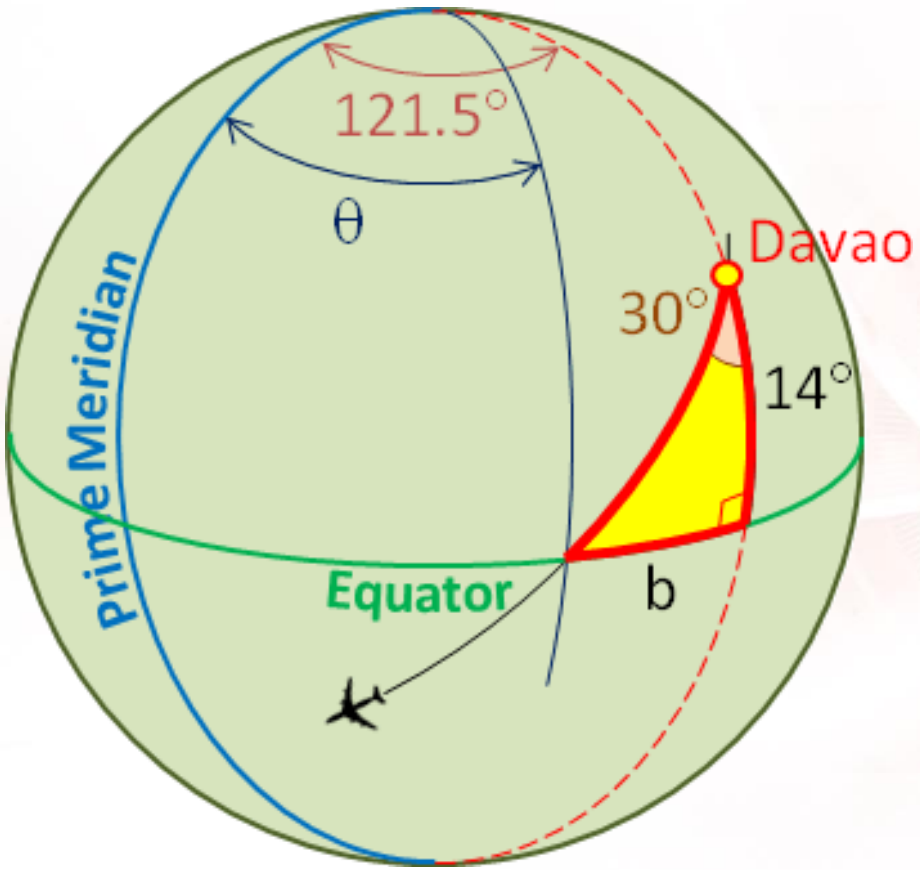
Reciprocal Identities

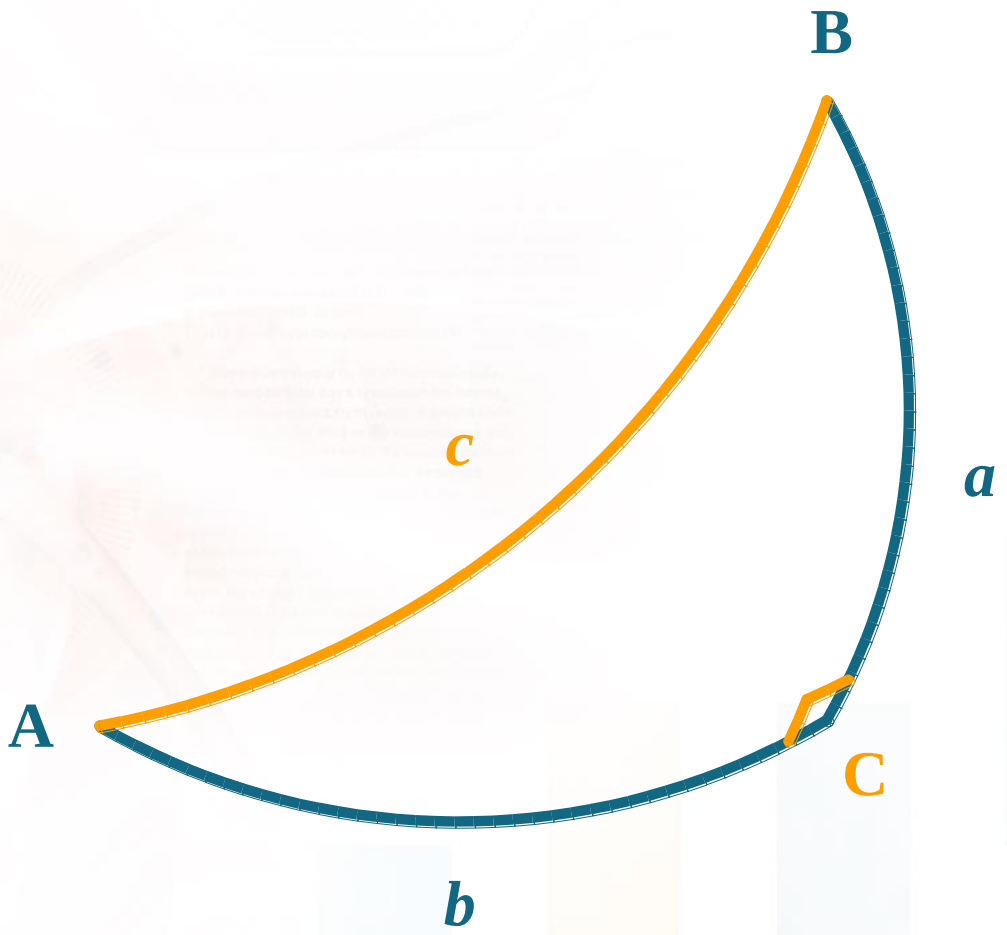
$$\csc(\theta) = \sec(90° - \theta)$$

$$\sec(\theta) = \csc(90° - \theta)$$

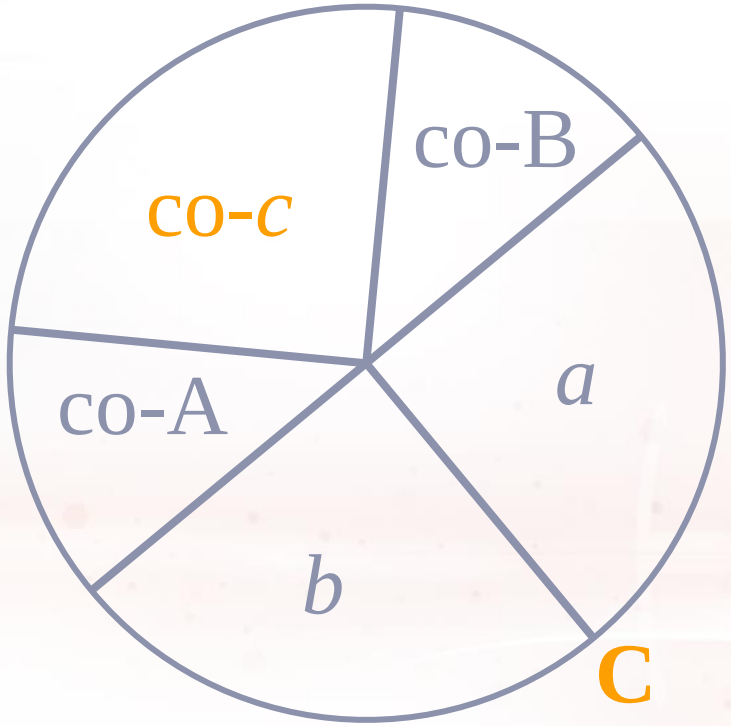
$$\cot(\theta) = \tan(90° - \theta)$$

NAPIER'S RULE

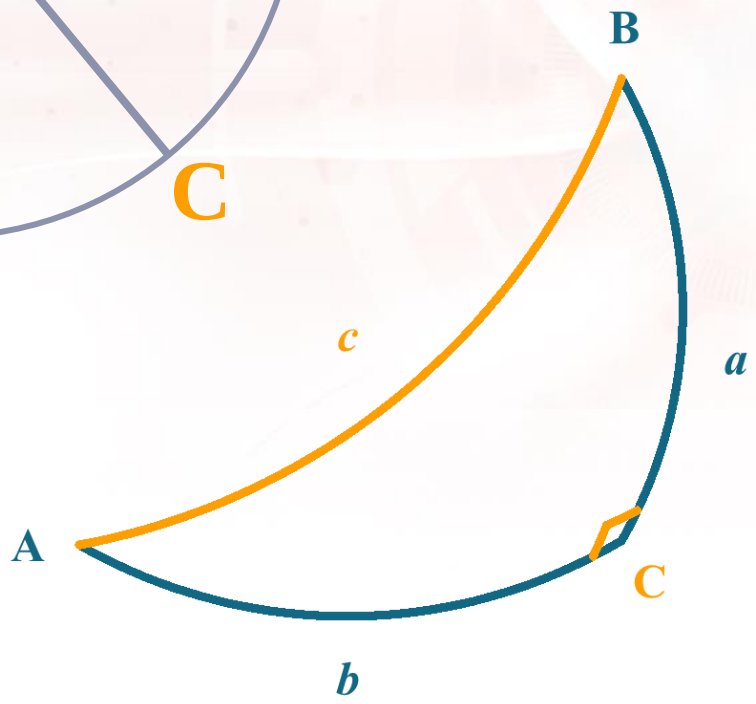

121.5°
θ
Davao
30°
14°
Prime Meridian
Equator
b


B
c
a
A
C
b

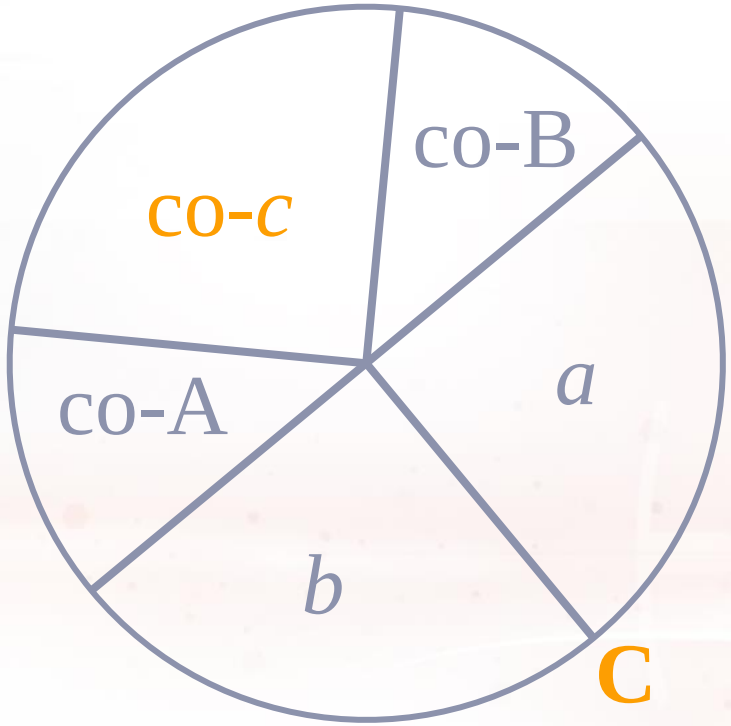
NAPIER'S RULE 1



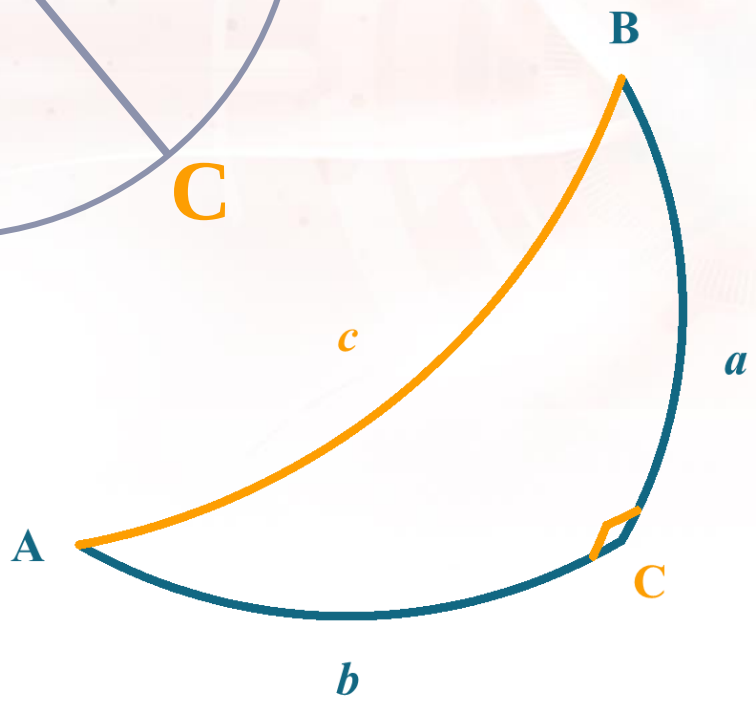
The **sine** of any **middle** part is equal to the product of the **tAngents** of the **Adjacent** parts

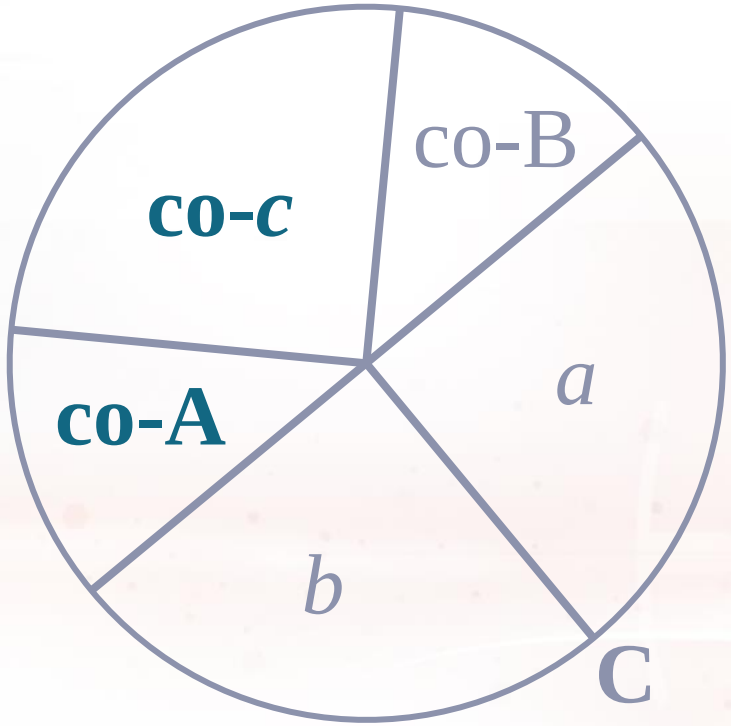


NAPIER'S RULE 2



The **sine** of any **middle** part is equal to the product of the **cOsines** of the **Opposite** parts



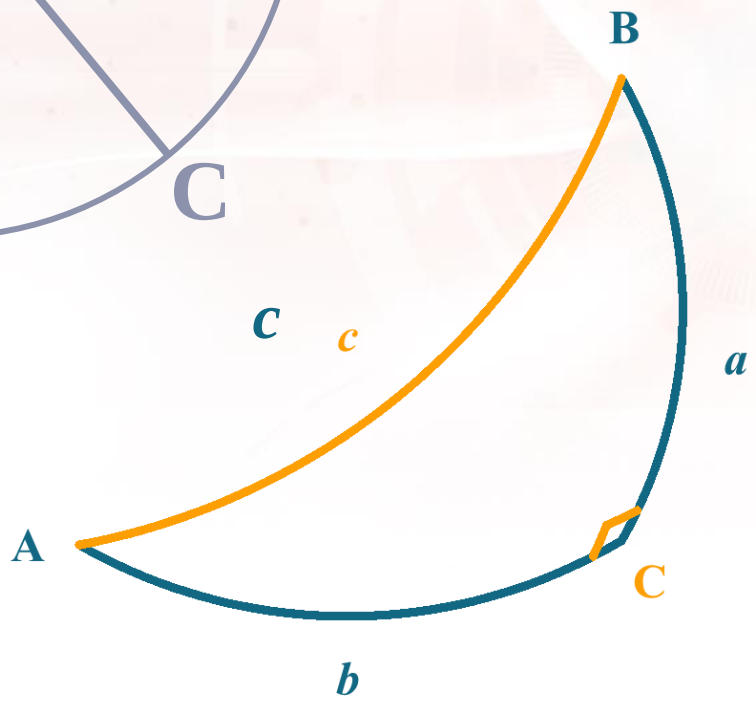


In a right spherical triangle ($C = 90°$)

$A = 69°50.8'$

$c = 72°15.4'$

$find\ B, a, b$

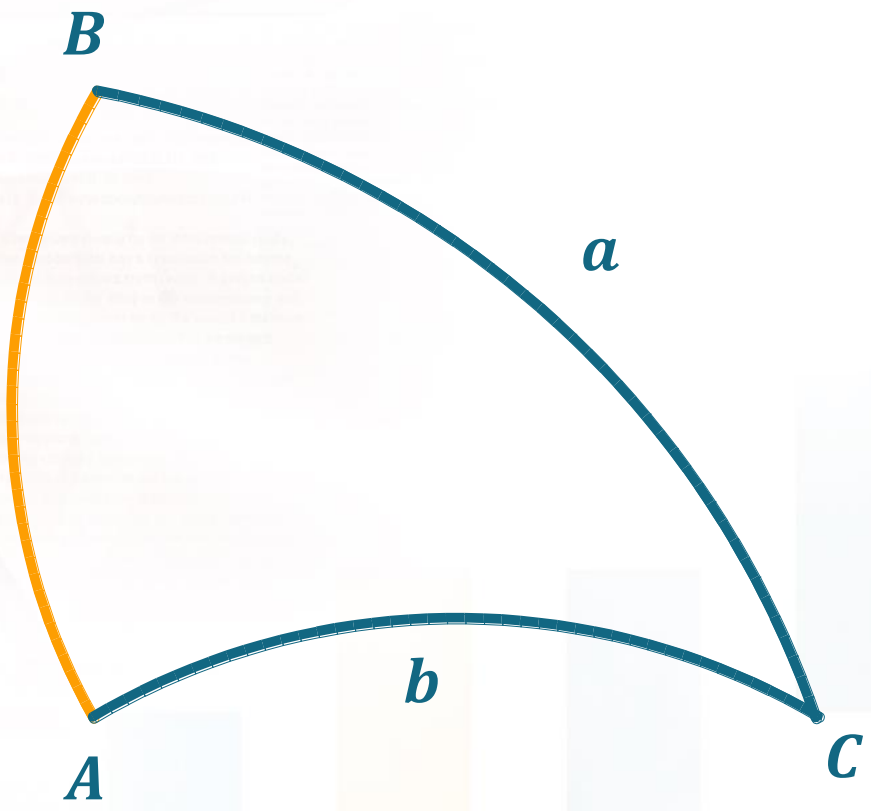


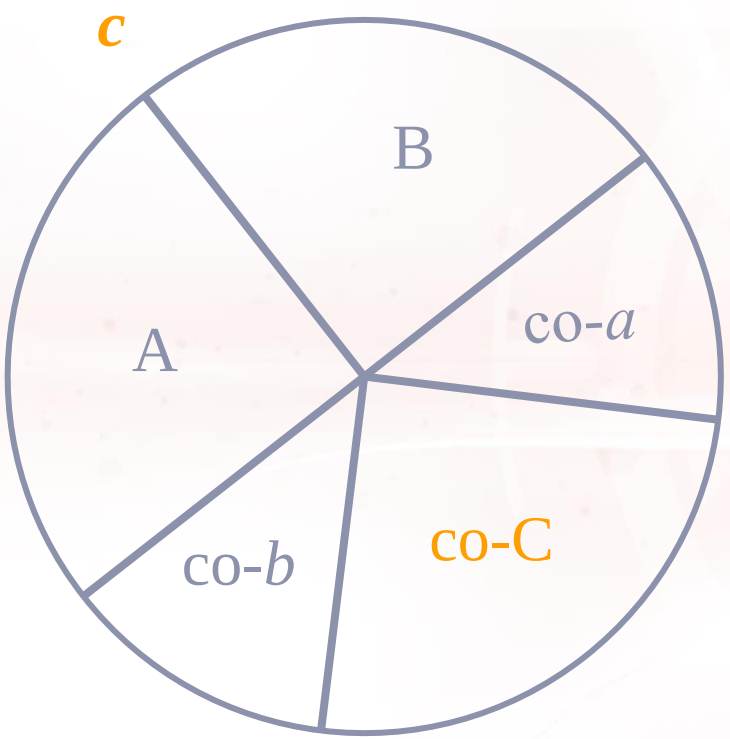
QUADRANTAL TRIANGLES

QUADRANTAL TRIANGLE

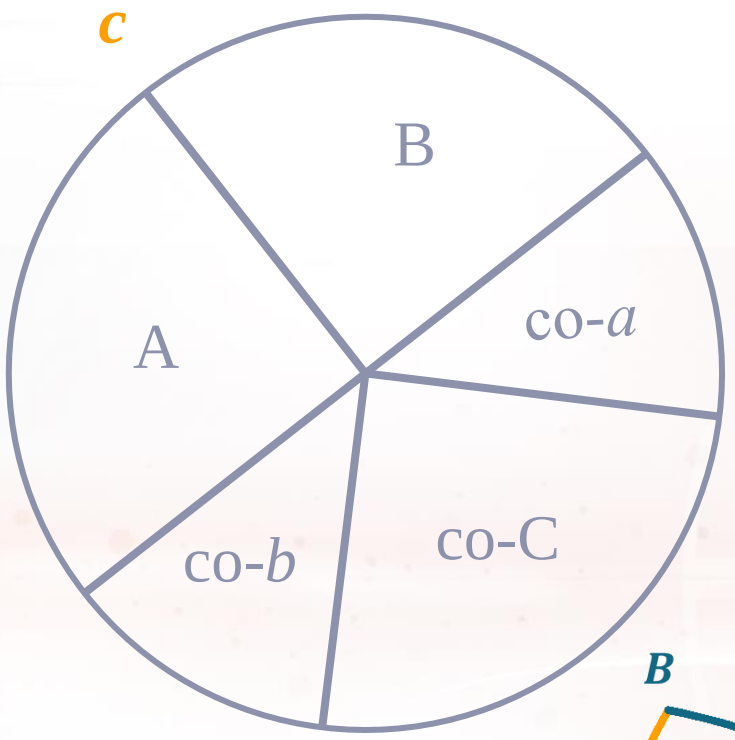
A spherical triangle having a **side** equal to **90°**

$c = 90°$





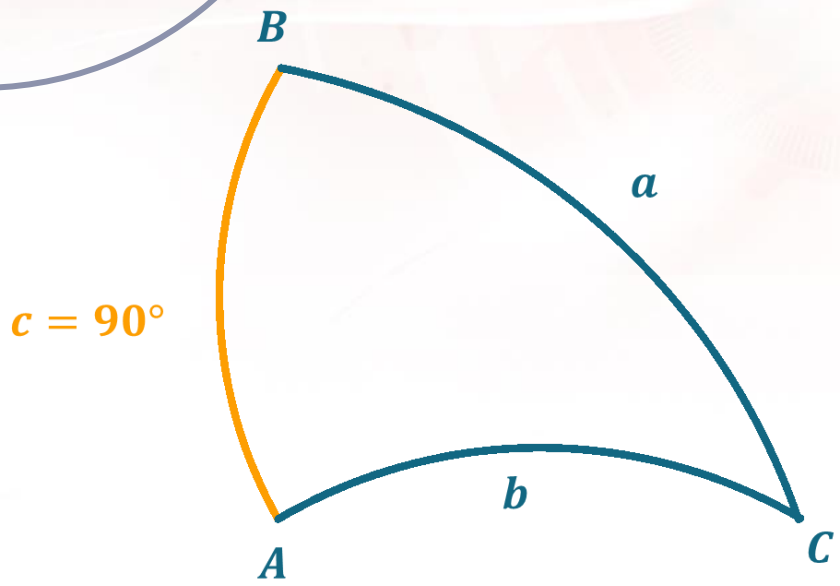
If the adjacent or opposite parts are **both angles**, or **both sides**, then an extra "**minus**" sign must be introduced



Solve the **quadrantal** triangle ($c = 90°$):

$a = 70°\ 7.8'$

$b = 52°\ 36.7'$



SEATWORK